



EarStudio ES100 : AK4375a DAC Digital Filter

Application Note

NOVEMBER 2018

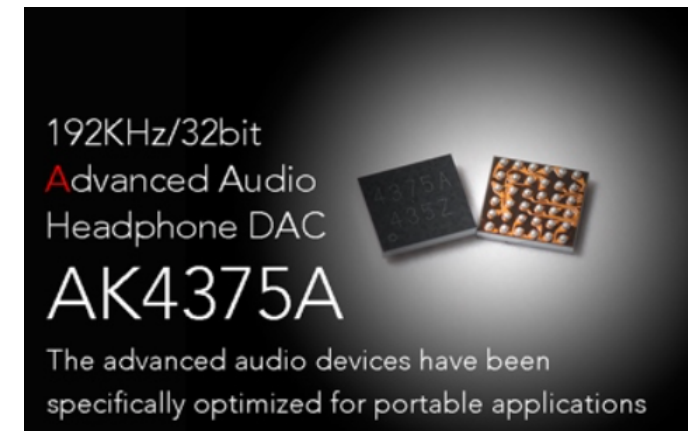
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ES100 AK4375a DAC Digital Filter

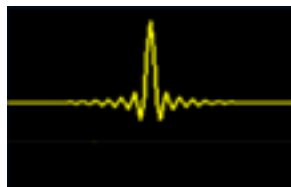
- Measurement

- ES100 Balanced Output (USB DAC mode)
- Measured with Audio Precision AP SYS-2722
- **Test Signal** : -6dB Rectangular Square Wave 1KHz



- Filter Coefficient: Summary

Traditional Sound

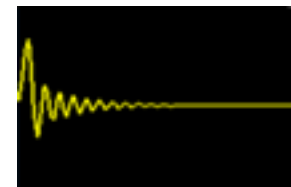


Sharp Roll-Off

Pre and post echoes / **powerful**

Slow edge

Acoustic Sound

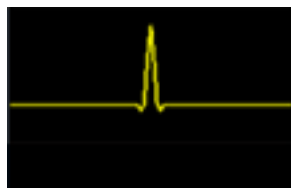


Short Delay Sharp Roll-Off

Post echo / **bass** sound

Sharp edge

Traditional Tone



Slow Roll-Off

Minimal echo / **original** sound

Middle edge

Acoustic Tone

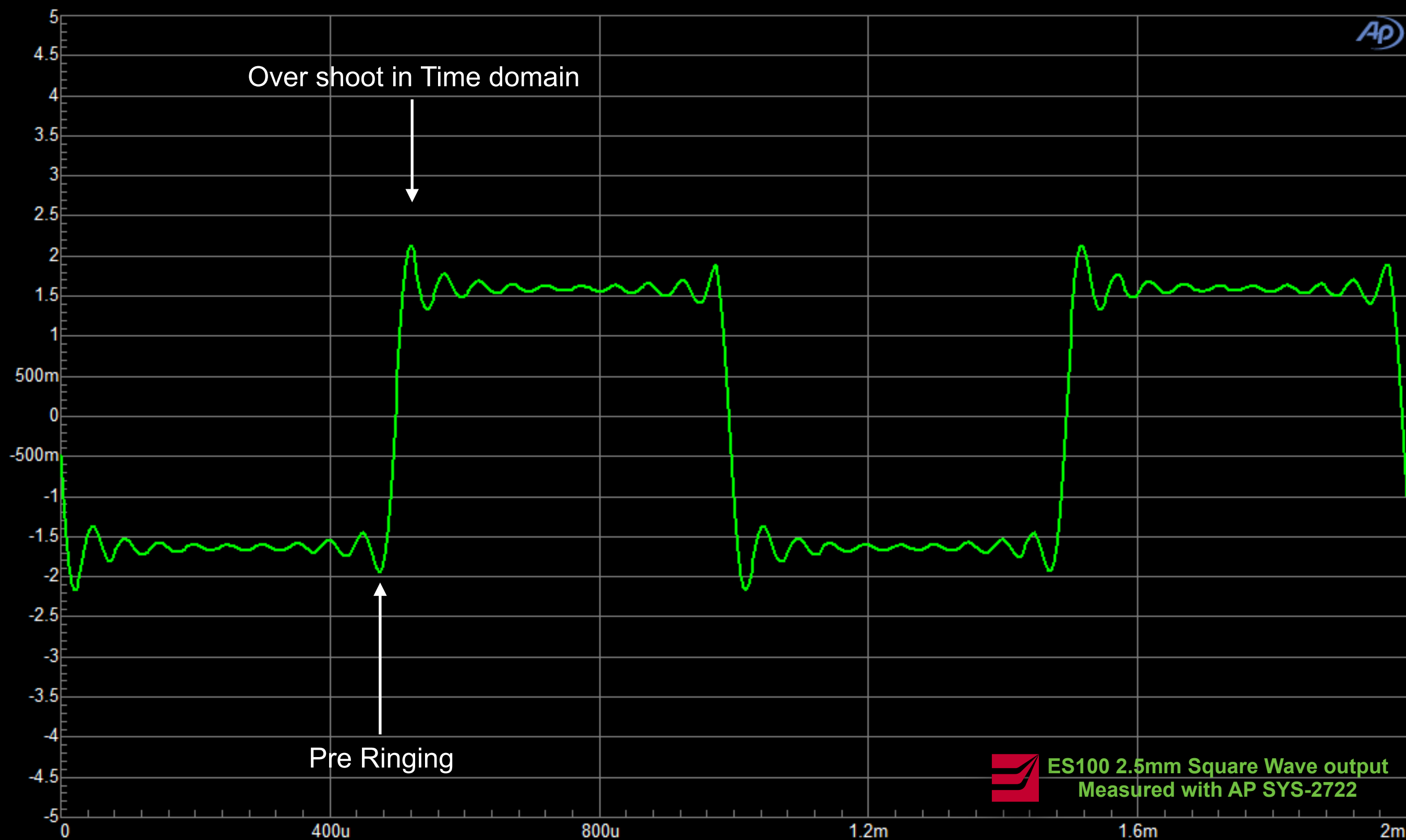


Short Delay Slow Roll-Off

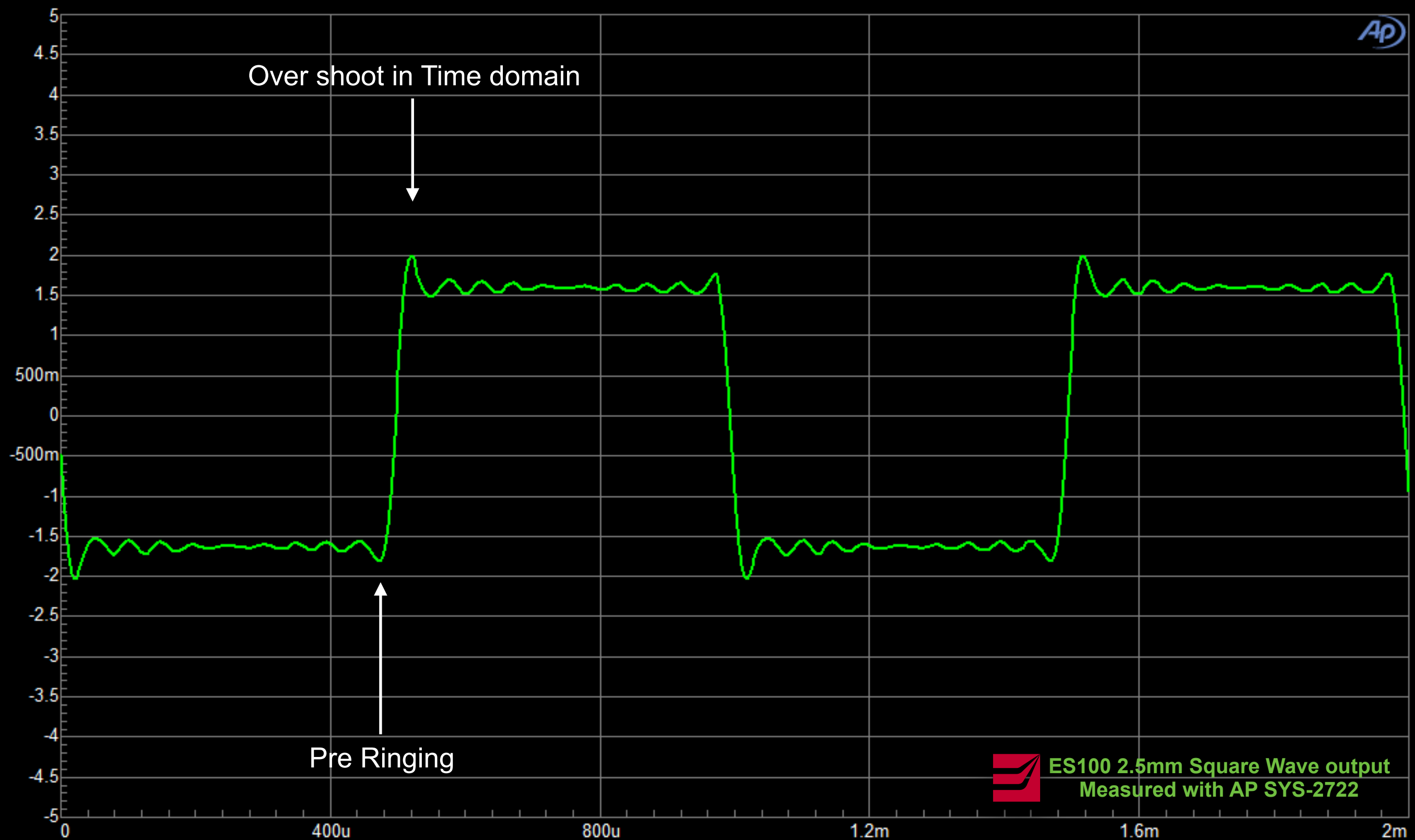
Minimal echo / **original** sound

Sharp edge

Square Wave through Sharp Roll-Off filter



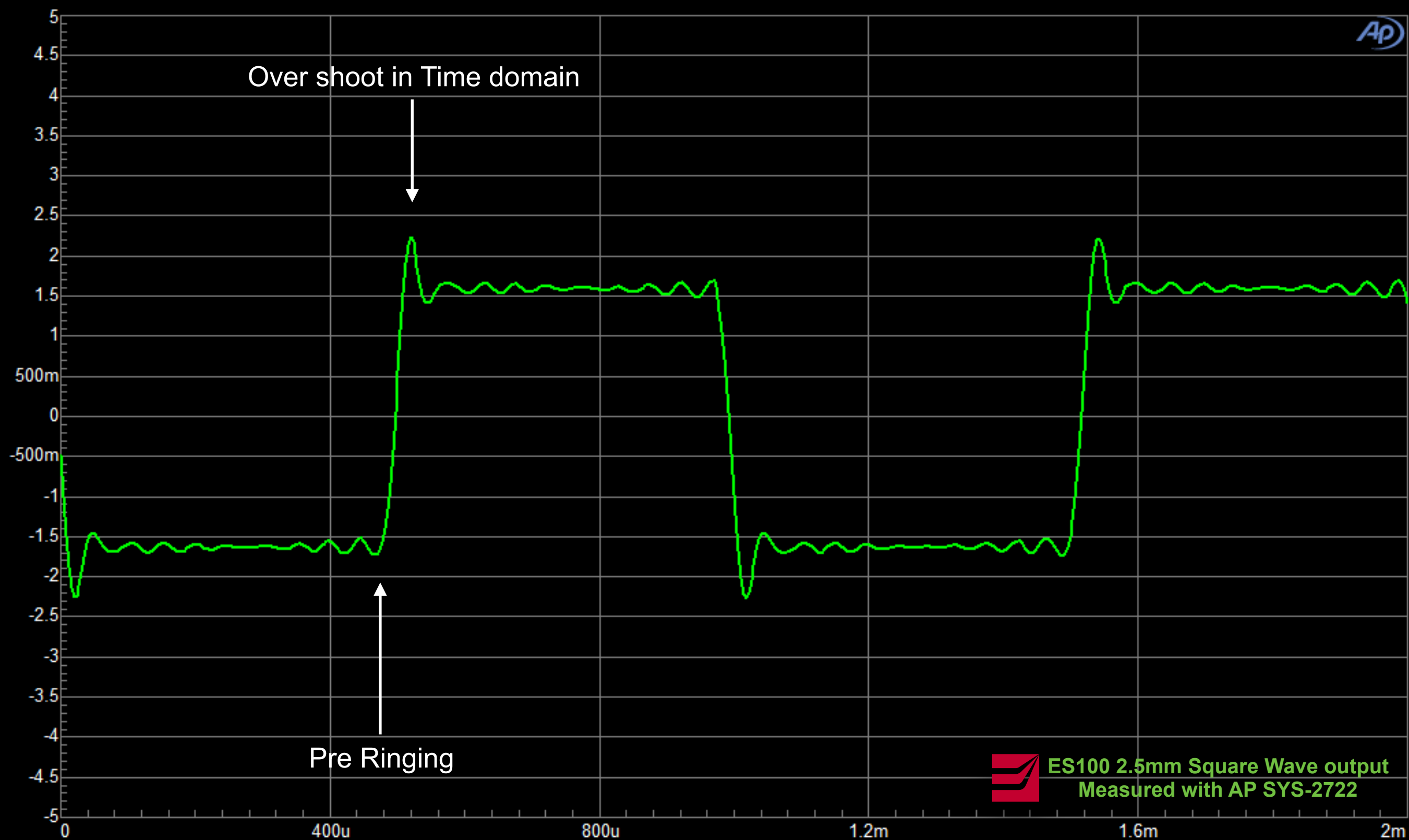
Square Wave through Slow Roll-Off



Square Wave through Short Delay Sharp Roll-Off

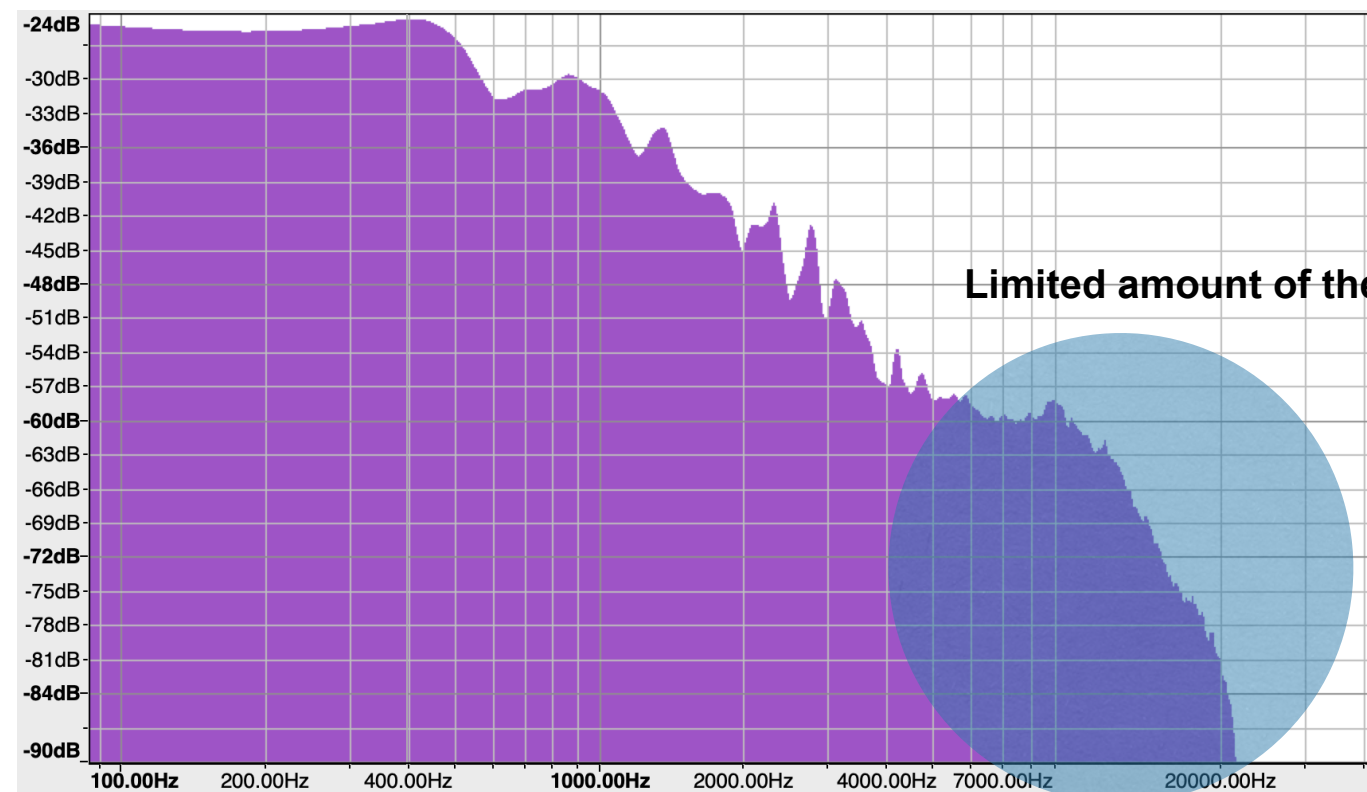


Square Wave through Short Delay Slow Roll-Off



Summary

- **Low Pass Digital Filter** is an essential functional block in **Delta Sigma D/A converter**.
- **The time domain characteristic of the digital filter coefficient** is more important than the frequency response of the filter.
- **Do NOT worry about the aliasing caused by the slow roll-off of the low pass filter.**
 - Yes, there would be the aliasing if you play a high-frequency and high-level tone or chirp signal.
 - But it's not designed to listen to those test signal.
 - The normal audio track other than those test signal just has the limited amount of the high-frequency terms.
- **Select and set the filter you want without worrying about the aliasing.**



Normal Audio Track Frequency Response